

Reading structured data

Project: Python readiness portfolio

1. Your mission

Your learning target

I can process a list of records consistently.

I can compute a total from named fields.

I can preserve record meaning while tracing a loop.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Process a list of records consistently

Compute a total from named fields

Preserve record meaning while tracing a loop

Retrieve what you already know

Which key stores the occupied count in {"capacity": 8, "used": 3}?

My starting explanation

This week's end goal

Trace reading structured data and explain the changed program before running it.

2. Notice the evidence

Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces. Treat the records as a dataset, and trace one record at a time before trusting a total.

List the given facts and one fact that is not supplied.

3. See how it works

```
rows = [{"capacity": 8, "used": 3},
        {"capacity": 6, "used": 6}]

total = 0

for row in rows:
    spare = row["capacity"] - row["used"]
    total = total + spare

print(total)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: First row

Evidence: $8 - 3 = 5$

Meaning: total becomes 5

Step or case: Second row

Evidence: $6 - 6 = 0$

Meaning: total stays 5

Read each row from left to right and connect the evidence to the stated rule.

The loop variable row refers to each record in turn. We calculate spare from that record, then add it. total starts before the loop. An empty list produces 0 because there are no updates; it does not tell us whether no cupboards exist or records were not collected.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

For spare contributions 2, 4 and 1, trace total starting at 0.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Add `{"count":4}` as a fourth row.

Prediction: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent field remains skipped.

5. Words that unlock the idea

Word · Meaning

Word: Dataset

Meaning: A collection of records

Word: Row

Meaning: One record in a table

Word: Field

Meaning: One named part of a record

Word: Accumulator

Meaning: A variable updated to keep a running result

Use two terms to explain a step in this week's example.

6. Follow a worked example

Add a third record with capacity 10 and used 4. Predict the total and list each contribution.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number

it produces.

A pictorial worked case: Reading structured data

Read rows with a missing-field policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

The facts we will use

Python 3 records: [{"count":2}, {"count":0}, {}]. Policy: include existing numeric counts, including zero; skip an absent field.

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Included values and absent field

	Count value	Include in mean?
Row 1	2	Yes
Row 2	0	Yes
Row 3	Absent	No

Read a cell using both its row and column labels.

Reading structured data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect three rows

Two rows contain count; the third does not.



2. Accumulate values

The included values sum to $2+0=2$, with denominator 2.



3. Report coverage and mean

Mean is 1; measured coverage is 2 of 3 rows.

CHANGE ONE THING

Add `{"count":4}` as a fourth row.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Inspect three rows

What happens and why: Two rows contain count; the third does not.

Step: 2. Accumulate values

What happens and why: The included values sum to $2+0=2$, with denominator 2.

Step: 3. Report coverage and mean

What happens and why: Mean is 1; measured coverage is 2 of 3 rows.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an observation.

What this example does not establish

This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

7. Find and repair a mistake

Putting $\text{total} = 0$ inside the loop discards earlier contributions. With the original rows it finishes at 0, the second contribution. Move the initialization before the loop and rerun both the original and reversed row order.

Locate the earliest unsupported step

A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

State one repair and the checks you would rerun.

8. Practise together

In the worked dataset, state the number of records and the spare spaces in the second record.

Add a third record with capacity 10 and used 4. Predict the total and list each contribution.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace all updates

Use workbook W2 for the connected example “Reading structured data”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth

row with count four.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

For spare contributions 2, 4 and 1, trace total starting at 0.

```
rows = [{"capacity": 8, "used": 3},
        {"capacity": 6, "used": 6}]

total = 0

for row in rows:

    spare = row["capacity"] - row["used"]

    total = total + spare

print(total)
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

In the worked dataset, state the number of records and the spare spaces in the second record.

A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

An empty dataset produces total 0. Can you conclude that every cupboard is full?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled reading structured data evidence sheet to Python readiness portfolio. Show how you can process a list of records consistently. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

12. Show what you know

This loop assumes every record has the required integer fields. A missing used field causes `KeyError`. Do not silently treat a missing count as zero; that would invent unused capacity.

An empty dataset produces total 0. Can you conclude that every cupboard is full?

An empty dataset produces total 0. Can you conclude that every cupboard is full?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Follow the evidence

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Read each row from left to right and connect the evidence to the stated rule.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Preserve records and report measured coverage

Trace the named fields and missing-value policy. Predict the output, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []

for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)

observed_total = sum(values)

if len(values) > 0:
    mean = observed_total / len(values)
else:
    mean = None

print(observed_total, len(values), len(rows), mean)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.